2 stage OTA design

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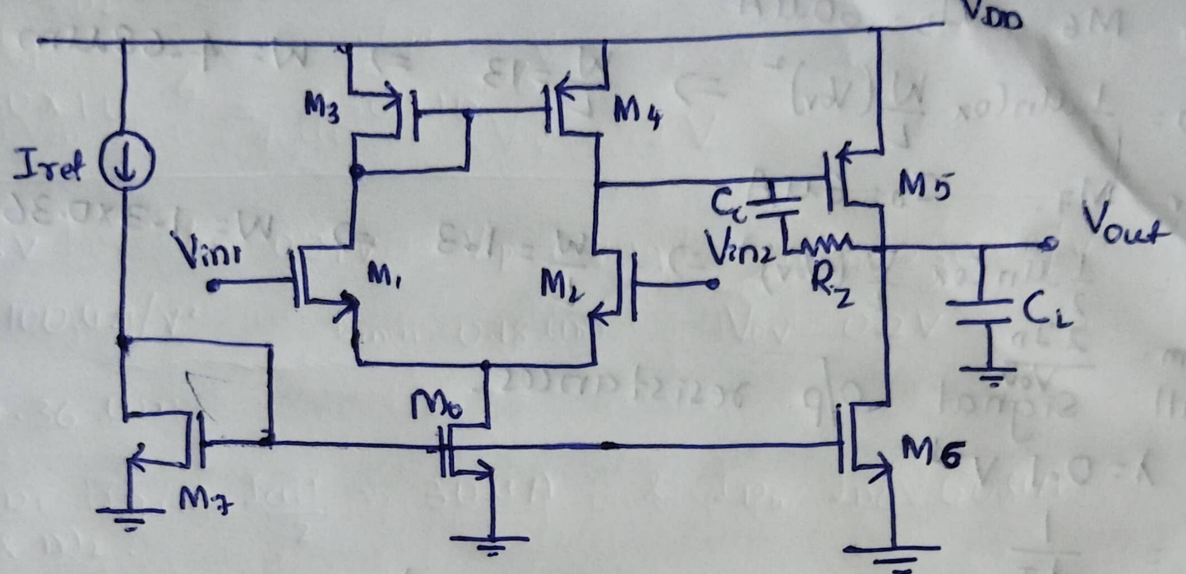


$V_{DD} = 1.8V$, $\mu_n C_{ox} = 230\,\mu A/V^2$, $\mu_p C_{ox} = 100\,\mu A/V^2$, $V_{ov} = 0.2V$

$V_{Tn} = 0.37V$, $V_{Tp} = 0.39V$. $L_{min} = 0.18\,\mu m$, $W_{min} = 0.27\,\mu m$

$C_L = 10pF$ — $C_c \geq 0.22\,C_L$, so, $C_c = 10pF$ & $R_z = 2K$

Let $L = 0.36\,\mu m$. & $I_{ref} = 6\,\mu A$

$I_{tail} = 10 \cdot I_{ref} = 60\,\mu A \Rightarrow I_{D_1} = I_{D_2} = 30\,\mu A$

and Also, take $I_{D5} = I_{D6} = 60\,\mu A$.

For M_1 & M_2 , $I_D = 30\,\mu A$

$$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 6.52 \Rightarrow W = 6.52 \times 0.36 = 2.35\,\mu m$$

$$g_m = \frac{2I_D}{V_{ov}} = 300\,\mu S$$

For M_0 , $I_D = 60\,\mu A$.

$$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 13.0 \Rightarrow W = 13 \times 0.36 = 4.68\,\mu m$$

$$g_m = \frac{2I_D}{V_{ov}} = 600\,\mu S.$$

For M_3 & M_4 , $I_D = 30\,\mu A$

$$I_D = \frac{1}{2}\mu_p C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 15 \Rightarrow W = 15 \times 0.36 = 5.4\,\mu m$$

$$g_m = \frac{2I_D}{V_{ov}} = 300\,\mu S$$

For M_5 , $I_D = 60\mu A$

$I_D = \frac{1}{2}\mu_p C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 30 \Rightarrow W = 30 \times 0.36 = 10.8\mu m$

$g_m = \frac{2I_D}{V_{ov}} = 600\,\mu S$

For M_6, $I_D = 60\mu A$.

$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 13 \Rightarrow W = 4.68\mu m$.

For M_7. $I_D = 6\mu A$

$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{ov})^2 \Rightarrow \frac{W}{L} = 1.3 \Rightarrow W = 1.3 \times 0.36 \simeq 0.47\,\mu m$

$g_m = \frac{2I_D}{V_{ov}}$

Small signal o/p resistances.

Let $\lambda = 0.1\,V^{-1}$

$r_o = \frac{1}{\lambda I_D}$

So, For $M_1, M_2, M_3, M_4 \Rightarrow r_o = \frac{1}{0.1 \times 30\mu A} \simeq 333\,k\Omega$

: For $M_5, M_6, M_0 \Rightarrow r_o = \frac{1}{0.1 \times 60\mu A} \simeq 167\,k\Omega$

For $M_7 \rightarrow r_o = \frac{1}{0.1 \times 6\mu A} = 1670\,k\Omega$

Theoretical gain.

Stage-1
$A_{v_1} = g_{m_1}(r_{o_1} \| r_{o_3}) = 300\mu \times 166k = 49.8$

Stage-2
$A_{v_2} = g_{m_2}(r_{o_5} \| r_{o_6}) = 600\mu \times 83k = 49.8$

Total gain

$A_v = A_{v_1} \times A_{v_2} = 49.8 \times 49.8 = 2480$

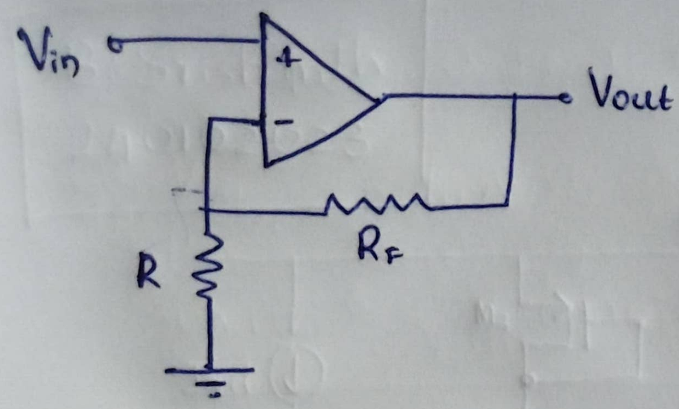
$A_v\,(\text{in dB}) = 20\log(2480) \simeq 67.89\,dB$

Power consumption:

$I_{total} = 60\mu + 60\mu = 120\mu A$

$P = 1.8 \times 120\mu = 216\,\mu W$.

closed loop.



Let $R = R_F = 100k$

Gain $= 1 + \frac{R}{R_F} = 2$

Gain(in dB) $= 20\log(2) = 6.02$ dB

From Experiment:-

(i) Open loop circuit configuration

- Experimental gain = 64.59 dB (at 1 kHz)
- Phase Margin = ~~180+121.871~~ = ~~58.129~~ 180 − 121.748 = 58.252°

(ii) Closed loop circuit configuration (for DC gain 2)

- Experimental gain = 6.015 dB
- phase margin = 180 − 101.107 = 78.893°

Table

	g_m	r_o	I_D	$V_{GS}-V_T$	L (in µm)	W (in µm)
M_0	600 µS	167 kΩ	60 µA	0.206	0.36	2.35
M_{1-2}	300 µS	333 kΩ	30 µA	0.23	0.36	4.68
M_{3-4}	300 µS	333 kΩ	30 µA	0.21	0.36	5.4
M_5	600 µS	167 kΩ	60 µA	0.21	0.36	10.8
M_6	600 µS	167 kΩ	60 µA	0.23	0.36	4.68
M_7	60 µS	1670 kΩ	6 µA	0.206	0.36	0.47